DYAL SINGH

Email- dyalsingh0987@gmail.com | LinkedIn- https://www.linkedin.com/in/dyalsingh2003/ | Madhya Pradesh (India) | Contact No- 8463833506 | GitHub- https://github.com/dyalsingh | Leetcode- https://leetcode.com/u/dyalsingh__/

PROFILE SUMMARY

Currently pursuing a Bachelor of Technology in Computer Science at RGPV (Hitkarini College of Engineering and Technology). Specialized in DevOps and AWS, engineering and automating cloud-native infrastructure to drive operational efficiency. Implemented, refined, and maintained deployment pipelines, reducing manual effort by 90% and accelerating deployment cycles by half.

TECHNICAL SKILLS

Containerization (**Kubernetes, ECR, EKS, Docker**) | CI/CD (**Gitlab, Jenkins**) **|** IAC (**Terraform**) | **MySQL** | **Monitoring** (CloudWatch) | **AWS** (EC2, VPC, S3, IAM, Lambda, DynamoDB, ECS, WAF, Billing alerts, etc.) | **Python** | **DSA with C++.**

PROJECTS

NODE.JS APPLICATION DEPLOYMENT USING KUBERNETES

- Deployed a Node.js TODO app on **K8s using KIND** and kubectl, **maintaining 99% simulated uptime**.
- Provisioned dedicated namespaces to ensure resource isolation and eliminate cross-environment conflicts.
- Implemented **rolling updates via Deployments and Services**, boosting deployment efficiency by 70%.
- Applied basic health checks and resource limits to enhance app reliability and minimize crash incidents.
- Automated deployments using YAML manifests and kubectl, **reducing manual intervention by 80%**.

AWS ECS (FARGATE): SERVERLESS DEPLOYMENT & MONITORING

- Streamlined workflows using AWS ECS (Fargate), **eliminating 90% of manual effort** and enabling rapid, serverless deployments.
- Containerized the application and managed image workflows via Amazon ECR, boosting deployment speed and consistency by 50% across environments.
- **Integrated Amazon CloudWatch** for centralized logging and real-time monitoring, accelerating **issue diagnosis by 30%.**
- Defined and enforced **IAM roles and policies to strengthen access controls**, ensuring zero unauthorized access incidents during testing.

ONLINE SHOPPING APPLICATION DEPLOYMENT

- Led the deployment of an Online Shopping application with a focus on automation, performance, and security.
- Scripted and engineered Docker installation via a **custom shell script** for consistent environment setup.
- Designed and implemented **multi-stage Docker builds**, **reducing image size by 1GB** and improving deployment speed by 50%.
- Configured '.dockerignore' and docker-compose for streamlined execution, health checks, and networking.
- **Utilized Docker Scout** to systematically detect and resolve build-time vulnerabilities.
- Released the application on Amazon EC2 and **Integrated the pipeline with GitLab CI/CD**

MOVIE RECOMMENDATION SYSTEM DOCKERIZED & DEPLOYED

- **Built** a Python-based movie recommendation system using a **supervised machine learning model** trained on labeled movie data, **leveraging a web API to access CSV files from TMDB**, to recommend 5 similar movies based on user search. *(College project)*
- **Packaged** and containerized the application using Docker to enable consistent, platform-independent deployments across development and production environments.
- **Released** the solution on AWS EC2 with automated setup, reducing configuration effort by 80%.

EDUCATION

BACHELOR'S DEGREE | Computer science (4th year) | Hitkarini college of engineering | **CGPA: 7** Jun/2022 – Jun/2026
Higher Secondary School | PCM & Computer | St. Gabriel's Senior Secondary School | **CGPA: 7.4** Feb/2021 – Mar/2022

INTERESTS

Fitness & Gym | Creative Interior Design | Architectural Sketching | Nature, sunsets, hiking